Data Preparation Pipeline for SCAR-MLKEM on Raspberry Pi 5

From native ML-KEM-768 measurements and conformance vectors to session-level, leakage-safe, model-ready records

01 · Measurement sources

Pi 5 native ML-KEM-768 responder
Controlled schedules, real decapsulations
One process executes one policy
Replication unit: the workload session

Vectors: NIST ACVP · Wycheproof
IoT-23: 8 CTU captures (context only)

No secret key material is persisted

02 · Integrity and schema audit

SHA-256 manifest for every raw file
Success flag and schedule digest per run
Same schedule across a session's policies

0 duplicates · 0 failed operations
0 time reversals · 0 budget violations

Times in ns; derived quantities in µs

03 · Workloads and seed namespaces

10 families: uniform, poisson, zipf,
bursty, periodic, nonstationary, abrupt,
thrashing, hotcold, varying_contexts
64 contexts · 1000 requests · 500 µs base

Disjoint seeds: train 10^5 · dev $2{\cdot}10^5$
dev_2 $4.1{\cdot}10^6$ · test $7{\cdot}10^6$–$7.4{\cdot}10^6$

New seeds checked against all earlier plans

04 · Leakage-safe splits

Outer unit: independent process session
Seed sets: train 60 · development 30 sessions
$\mathcal{S}_{\text{tr}} \cap \mathcal{S}_{\text{dev}} = \mathcal{S}_{\text{tr}} \cap \mathcal{S}_{\text{test}} = \mathcal{S}_{\text{dev}} \cap \mathcal{S}_{\text{test}} = \varnothing$

Inner folds: expanding forward blocks
Purge 70 ms = 50 ms history + 20 ms label

Test seeds generated only after the freeze

05 · Causal features and labels

11 causal features from the 50 ms history
$c_5, c_{10}, c_{20}, c_{50}$: counts in 5–50 ms windows
$\text{gap}_{50}, \bar{g}, \text{cv}_g, \text{exp}_{10}$, share shift, rates

$y_h = \mathbf{1}[\text{context reused within } h]$
$h \in \{5, 10, 20\}$ ms; primary 10 ms

Excluded: key id, family, session, outcome

06 · Scaling and records

$z = (x - \mu_{\text{tr}})/\sigma_{\text{tr}}$, fitted on the train fold
Logistic scaled; tree models unscaled

Per request: timings, hits, admissions,
stage timers, arena occupancy
Per process: thread CPU, RSS, lifecycle

Secret material never enters any record

07 · Frozen operation costs (Pi calibration)

$C_{00} = 31.83,\ C_{01} = 28.78,\ C_{10} = 22.77,\ C_{11} = 19.72$ µs
$a_P = 1.12,\ a_S = 0.36,\ e_P = 0.58,\ e_S = 0.41$ µs

$d_P = C_{00} - C_{10} = 9.06, \quad d_S = C_{00} - C_{01} = 3.05$ µs
$d_I = C_{10} + C_{01} - C_{00} - C_{11} = -0.002$ µs

C_{ps}: request cost with public (p) and secret (s) tiers retained
Measured on the Pi: medians of eight process means
Spread ≤ 0.19 µs; shared by every cost-aware policy

08 · Prepared data and configuration

Session frame: one row per process
Keys (family, seed, arena, setting) + CPU/request,
hit rates, p95/p99, peak RSS, late clears

Registered plans · frozen design · verified raw
Verified sample: 0 failed processes,
all required fields present

Ready for development and confirmation studies